

REORGANISATION OF AN OUTPATIENT DEPARTMENT IN A HOSPITAL

PROJECT REPORT – GROUP 23

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1 INTRODUCTION

1.1 Problem description

The construction of a cyclic appointment schedule stands as a pivotal yet difficult task in the optimization of a radiology department's efficiency. There are many challenges and important decisions that need to be addressed, each bearing weighty significance. The process involves many strategic and tactical decisions related to capacity distribution and the timing of time slots. In this paper, the difficulties inherent in the creation of such schedules will be investigated.

The cyclic appointment schedule consists of time slots dedicated to urgent or elective patients only. Urgent patients are patients that have entered the hospital without an appointment and should be treated on the same day. The aim of this study is to evaluate the effects of different scenarios on waiting times for urgent and elective patients while considering the impact on the daily overtime, using simulation techniques.

1.2 Objective

Currently, there are 14 time slots dedicated to urgent patients. The department has decided that the slots dedicated to urgent patients should range between 10 and 20. This has a significant impact on the waiting time of both elective and urgent patients.

The timing of time slots for urgent patients is currently installed at the end of each morning block. They are distributed evenly over the morning and afternoon blocks and over the days. This represents the first strategy. The second strategy proposes an even distribution of urgent time slots over the day taking the opening hours of the department into account. The third strategy is planning an urgent time slot each time after a block of six time slots for elective patients.

On the operational level, there are some important decisions that need to be made in order to determine to performance of the schedule accurately. To evaluate the cyclic appointment schedule, four performance measures are used: the appointment waiting time of elective patients, the scan waiting time of elective patients, the scan waiting time of urgent patients and the overtime required in the department to examine all patients.

There is a focus upon the appointment waiting time of elective patients and the scan waiting time of urgent patients, as these seem the most important objectives when determining the capacity distribution and the timing of the time slots for urgent patients. To find a suitable objective function, the relevant waiting times are by multiplied by the maximum desired waiting time postulated by the management for the different patient types. Resulting in the following objective function:

$$\begin{array}{ll} \text{Minimize} & w_e * Wt_e + w_u * Wt_u \\ \text{With} & w_i \quad \text{Priority weight of patient type } i \\ & Wt_i \quad \text{Average waiting type for patient } i \end{array}$$

Another critical operational decision is related to the appointment scheduling rules. To capture appointments, the hospital currently uses a First Come First Served system, but there are alternatives to this: the Bailey-Welch rule, the blocking rule, and the benchmarking rule. Further explanation of these rules can be found in Appendix [7.1](#).

1.3 Research questions and hypotheses

1.3.1 Research question

Given the above-described problem and objectives, the following research question can be articulated as follows:

“What is the optimal cyclic appointment schedule for an outpatient radiology department with a single server that minimizes the scan waiting time for urgent patients and appointment waiting time for elective patients while also considering the impact on daily overtime and the scan waiting times for elective patients?”

1.3.2 Hypotheses

Hypothesis

H_0 : None of the other examples do better than the baseline scenario.

H_1 : At least one of the other examples does better than the baseline scenario.

Subhypotheses:

- Subhypothesis 1:
 - H_0 : the usage of a different strategy for the timing of the time slots does not improve the objective function by reducing the objective value
 - H_1 : the usage of a different strategy for the timing of the time slots does improve the objective function by reducing the objective value
- Subhypothesis 2:
 - H_0 : the usage of a different appointment scheduling rule does not improve the objective function by reducing the objective value
 - H_1 : the usage of a different appointment scheduling rule does improve the objective function by reducing the objective value
- Subhypothesis 3:
 - H_0 : the usage of a different number of time slots for urgent patients does not improve the objective function by reducing the objective value
 - H_1 : the usage of a different number of time slots for urgent patients does not improve the objective function by reducing the objective value

2 INPUT DATA

Before delving into the issue, the analysis begins by characterizing the available data to gain a better understanding. Variability in both patient arrivals and service times also influences the outpatient department's performance. The variability in when elective patients arrive is related to both the time when they make their phone call to schedule the appointment and their actual arrival time.

The phone call time of these non-urgent patients follows a memoryless negative exponential distribution, while the number of arrivals per day is following a Poisson distribution. The tardiness distribution function is modelled by a normal distribution with a mean of 0, which means that the patients are expected to arrive on time with a standard deviation of 2,5 minutes. There is also a 2% probability of not showing up. The urgent patient arrivals are modelled by a negative exponential distribution with different rates depending on the opening hours of the department.

Service times also show variability that affects appointment and scan waiting times. Elective scan durations follow a normal distribution with a mean of 15 minutes and a standard deviation of 3 minutes. There are no significant differences between the distributions of different scan types. This differs from urgent patients, whose scan durations vary based on the type of scan, with each type estimated to follow a normal distribution.

Alternative configurations will be used to make the best interrelated decision regarding to the capacity allocation between patient types, the allocation of time slots throughout the cyclic horizon and the appointment scheduling rule. The input data, decision variables, and their details are available in [Table A](#), which is attached to the appendix.

3 SIMULATION MODEL

3.1 Type of simulation

Discrete event simulation is intended to model an operation/a system as a sequence of time where events occur at specific, separable instances in that time. Each event marks a change of state in the system. Between 2 consecutive events, it is assumed that no change is expected to occur. This means that the simulation time can directly jump to the occurrence time of the next event. This process is called next-event time progression. An example can be found in the appendix at [7.2](#) (Robinson, 2004)

3.2 State descriptor

The state descriptor or the system state is the representation of the system at a particular time. The system can be represented by a single element or a vector. This vector can consist of different discrete random variables.

A state descriptor normally consists of three concepts. The first concept is the performance measure. The performance measure in the state descriptor is represented by the different waiting times and status of patients in the system, such as the number of patients waiting for an appointment, the number waiting for a scan, and so on. These measurements help evaluate the efficiency of the system. The second concept is uncertainty. The fact that a distinction is made between elective and urgent patients, and the fact that both types follow a different distribution (Poisson for elective and exponential for urgent), creates a certain degree of uncertainty. Third, there are the system-specific variables. The fact that there are different patient types in the system already gives a hint of heterogeneity. The fact that different strategies and rules are considered also adds some further system complexity.

Taking these matters into account, this state descriptor is obtained:

- The number of elective patients waiting for an appointment
- The number of elective patients waiting for a scan
- The number of elective patients in scan
- The number of urgent patients waiting for a scan
- The number of urgent patients in scan

3.3 Simulation time

Reporting on the simulation time more specifically execution time, helps assess the efficiency of the simulation and model complexity. On top of that it allows for further insights into the scalability of the simulation. It depends on the hardware used (here: Lenovo legion: Intel(R) Core(TM) i7-9750H CPU @ 2.60GHz 2.59 GHz, with 16GB of RAM) and simulation software (here: Python). This execution time is 90 seconds for the final 13 runs of the simulation for 500 weeks simulation length including the variance reduction method.

On top of that, simulation time can also refer to the tracked times, within the simulation. There is an internal clock keeping track of the arrivals on each of the days. Depending on the day, the opening hours differ and on Sunday the department will be closed. The internal clock is thus expressed by the hours, days and weeks passing.

3.4 Program structure

The program starts with creating a constructor, that is used to instantiate a simulation, where it defines different arguments. Firstly, specifying the 'filename' parameter, denoting the name of the input file housing the schedule data. Following this, defining 'W' as the integer representing the number of weeks to simulate, determining the runtime extent of the simulation. Similarly, 'R' is designated to represent the number of simulation runs, a key factor in assessing the robustness and reliability of the simulated outcomes. Lastly, the 'rule' parameter is established to dictate the specific appointment scheduling rule to be applied throughout the simulation.

The construct is then used to call the `runSimulation()` function, this function acts as the main organizer of the simulation, directing all its steps. The `runSimulation()` function includes several key functions. Firstly, `setWeekSchedule()` initializes the weekly schedule based on the provided input file and applied rules. Subsequently, `resetSystem()` ensures a clean slate for each replication by resetting all relevant variables. Lastly, there is the `runOneSimulation()` function.

The core simulation iteration is executed through the `runOneSimulation()`. Within this method, patient generation occurs via `generatePatients()`, where new patients are created considering parameters such as arrival time, tardiness, and procedure duration. Patient scheduling is carried out in `schedulePatients()`, determining the next available time slot for each patient based on the current schedule and applying specified rules. Sorting of patients based on appointment time and arrival time is facilitated by sorting functions. Finally, waiting time calculation and output metrics calculation are performed, providing insights into average waiting times and overtime for the entire simulation iteration. Note that in `runOneSimulation()` and `schedulePatients()`, the code that was provided in the assignment was altered. This was to ensure the correct calculation for running averages over the weeks. Originally an average was taken every week, simply taking the number of patients per week into account. This way the running averages were merely approximations of the variables needed to calculate the objective values. The alteration made sure that the waiting and overtime averages across the weeks were based on the total number of patients, keeping track of the number of patients in the weeks that had already passed. This ensures that each patient in the whole simulation contributed equally to the average, which originally was not the case.

During the simulations, various output metrics are generated and written down in a CSV file named 'output.csv'. This file serves as a record of the simulation results, containing data such as the average waiting times for elective appointments (Avg_Elective_App_WT), elective scans (Avg_Elective_Scan_WT), and urgent scans (Avg_Urgent_Scan_WT), as well as the average overtime (Avg_OT) and objective value (Objective_Value) for each run.

There are 3 distinct codes provided in the appendix, because some parts of the analysis required specific changes. [The first file](#) contains the code to extract information out of one simulation run. [The second code file](#) applies the chosen variance reduction method, this was a crucial step that made it able to accurately perform an analysis in the experimental design. [The third code](#) is a simplified version of the third for the one factor at a time approach.

4 SIMULATION RUNS

4.1 Baseline model

In this simulation study, the baseline model is given by the current structure of the radiology department. The hospital currently uses 14 urgent patient slots, schedules according to the plain FCFS (First Come First Served) rule and employs capacity distribution strategy 1.

The objective function is decided by combining the elective appointment waiting times and the urgent scan waiting times for patients. The averages of these parameters are shown in Figure 1.

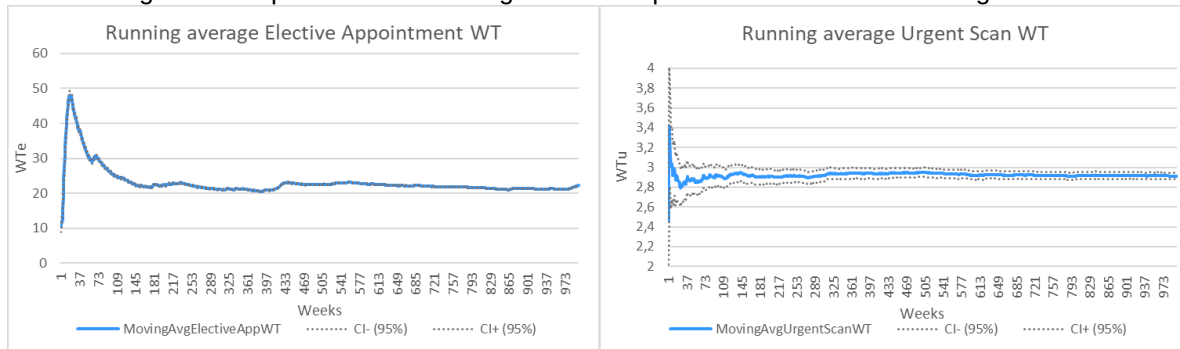


Figure 1: 1(a) (left) represents the running average for the elective appointment waiting time over the simulation period (in weeks), 1(b) (right) represents the running average for the urgent scan waiting time over the simulation period (in weeks)

The 95% ($\alpha=0.05$) confidence intervals were calculated using the following formulas:

Before the 30th patient, using the student t distribution: $\left[\bar{X} - \frac{t_{n-1, \frac{\alpha}{2}} \cdot \sigma}{\sqrt{n}}; \bar{X} + \frac{t_{n-1, \frac{\alpha}{2}} \cdot \sigma}{\sqrt{n}} \right]$

After the 30th patient, using the normal distribution: $\left[\bar{X} - \frac{z_{\frac{\alpha}{2}} \cdot \sigma}{\sqrt{n}}; \bar{X} + \frac{z_{\frac{\alpha}{2}} \cdot \sigma}{\sqrt{n}} \right]$

These formulas are used for all the confidence intervals where it is not specified differently.

4.2 Length of a simulation run

The length of each simulation run was determined using, what our group called, a 'pre-screening design', based on the one-factor-at-a-time approach. The simulation was run for several factors (discussed in the hypotheses), to get a proper view on how different setups converge to steady-state. Note that steady-state behavior is detected visually, furthermore the principle of common random numbers was not used here. This was to see whether differently generated random numbers would result in different simulation lengths.

First, the different scheduling rules were altered for the baseline model, keeping the number of urgent slots and capacity distribution strategies the same. Figure 2 shows visually that the time (in weeks) until the system reaches steady state does not change for the different rules. It was thus decided that altering the rules would not affect the needed simulation length. The choice of simulation length in one

situation would thus be appropriate for all the rules. See [Table C](#) in the appendix for the plots of the corresponding running averages for elective appointment waiting time and urgent scan waiting time.

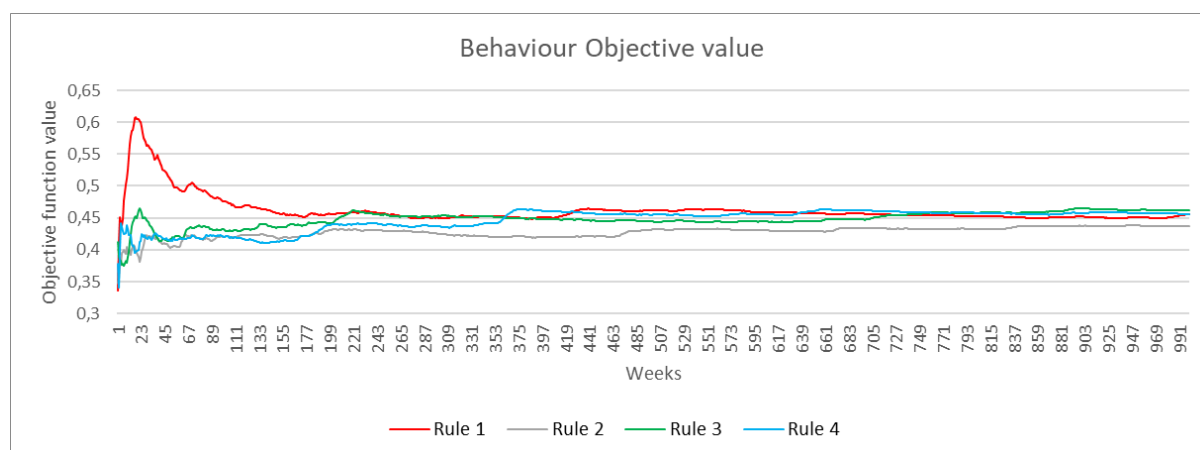


Figure 2: This is a combination of the running averages of the objective values for different rules, for the baseline model (14 urgent slots, strategy 1)

Subsequently, the capacity distribution strategies for urgent time slots get compared. Keeping the rules and number of slots the same. Figure 3 shows that the strategy does not have a significant impact on the choice of simulation length. [Table D](#) in the appendix shows the corresponding relevant plots.

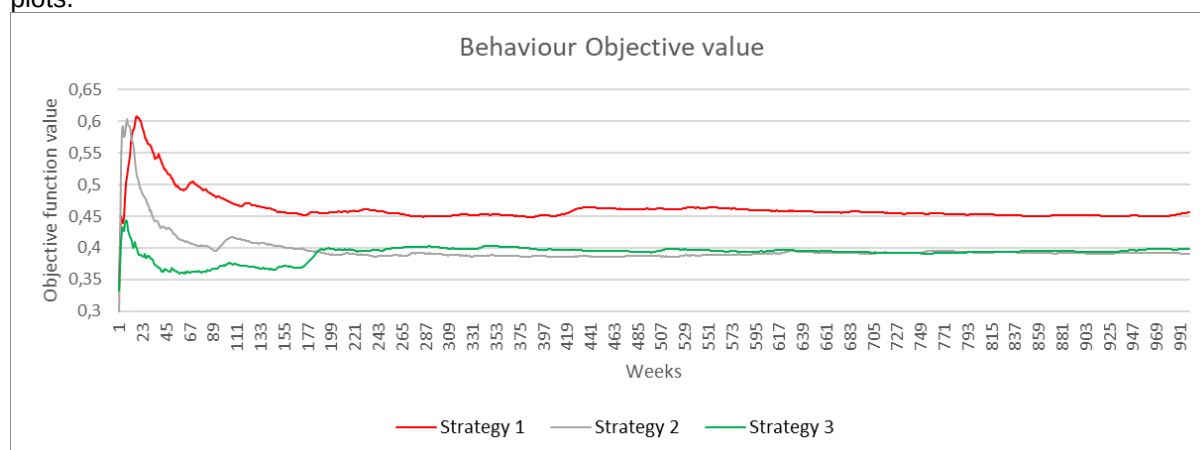


Figure 3: This is a combination of the running averages of the objective values for different strategies, for the baseline model (14 urgent slots, rule 1)

After that, the number of urgent slots in the week schedule was changed. Figure 4 shows that no significant difference in number of weeks was needed to reach steady state for the objective value. Combining these three findings led to a choice of 500 weeks as simulation run length ensuring that the sample path is in a steady state situation when the simulation run finishes. [Table E](#) in the appendix shows the corresponding relevant plots.

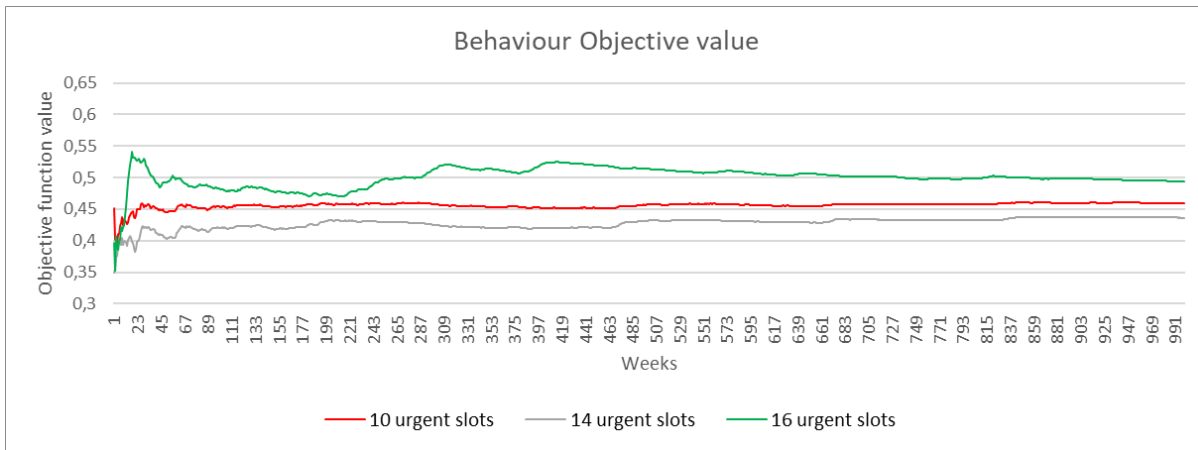


Figure 4: This is a combination of the running averages of the objective values for different amounts of slots, for the baseline model (strategy 1, rule 2)

4.3 Warm-up period

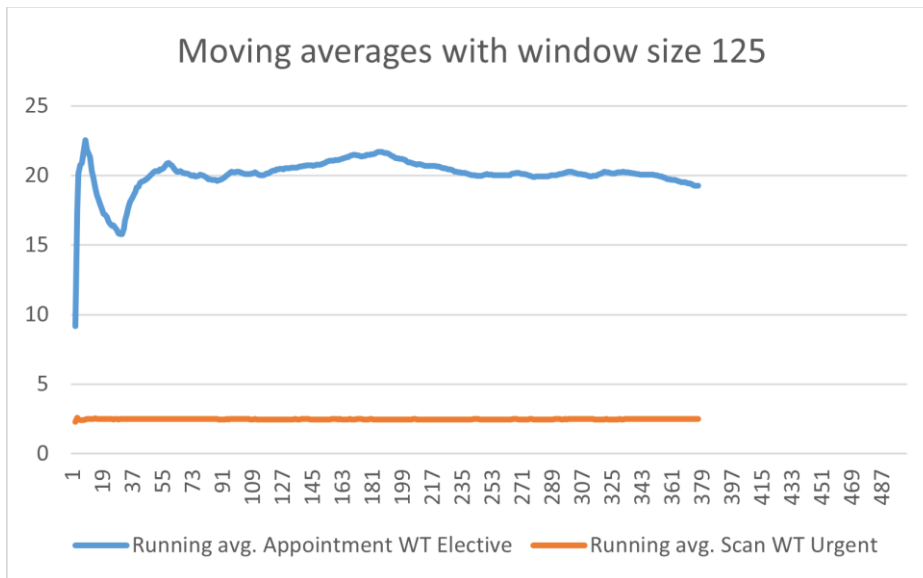


Figure 5: Plot showing the running averages of both the appointment waiting time for elective patients and the scan waiting time for urgent patients with window size 125 to determine the warm-up period

To determine the warm-up period in our simulation model of the outpatient radiology department, the Welch algorithm was employed. This method relies on calculating the moving average of the output statistics and identifying the point in the time series where the moving average flattens out, indicating the warm-up period. In this analysis, it was applied to 3 different strategies and then the longest warm-up period was taken as the most conservative choice.

1. Using a simulation run length of 1000 weeks (T), time series data are obtained for the output variable Y , denoted as Y_{it} .
2. The mean of the output data in each period is calculated, denoted as $\bar{Y}_t$, using the formula:

$$\bar{Y}_t = \frac{1}{n} * \sum_{i=1}^n \bar{Y}_{it}$$

where n represents the number of observations.

3. Next, the moving average for $\bar{Y}_t$ is computed using a specified window size, w .

4. The moving average $\bar{Y}_t(w)$ is plotted as a time series using the following formulas:

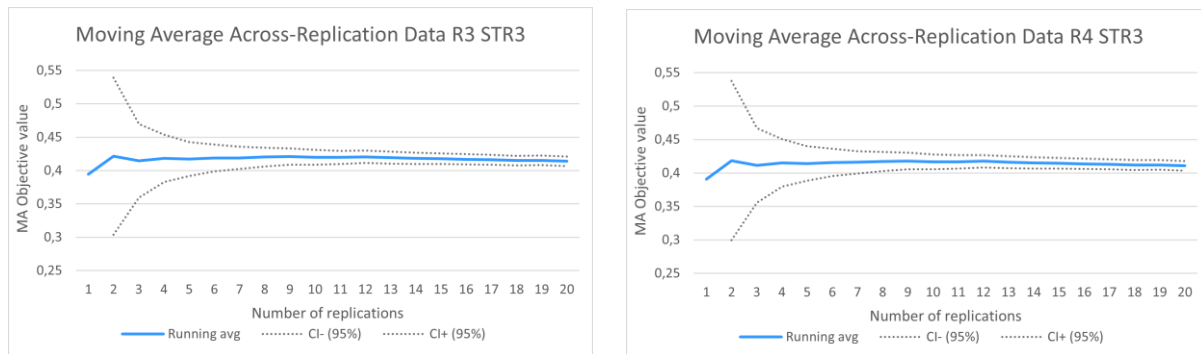
$$\bar{Y}_t(w) = \sum_{\tau=1}^{2t-1} \frac{\bar{Y}_\tau}{2t-1} \text{ if } t = 1, \dots, w \text{ and}$$

$$\bar{Y}_t(w) = \sum_{\tau=0}^{2w} \frac{\bar{Y}_{t-w+\tau}}{2w+1} \text{ if } t = w+1, \dots, T-w$$

According to the Welch algorithm, one proceeds iteratively, incrementing the window size w until the plotted graph appears smooth. The plotting of the graph started with a window size of 125, which then became the final choice, as the time series of the moving average with that window size showed a flat trend after observation. Based on this, a warm-up period of 60 weeks was selected, meaning that the first 60 weeks of each simulation run are excluded because these initial deviations are correlated and dependent on the initial state.

4.4 Number of simulation runs

In this report, the experimental design utilizes the replication method, crucial for obtaining objective estimates in subsequent analyses. To determine the number of simulation runs, there has been made use of the replication data for the objective function value. For the screening design, 7 replications suffice as they generally satisfy the half-width criterion of 0.02. Similarly, for the comparative design, 13 replications are chosen, meeting the typical half-width criterion of 0.01. The half-width represents the maximum allowable error. These choices have been made based on moving averages from the across-replication data, using the different rules and strategies.



Figures 6a and 6b: overview across-replication data R3&4 STR3

On the vertical axis of Figures 6a & 6b the moving average of the objective value can be found. The horizontal axis shows the number of replications. This visualizes that generally the half width of 0.02 and 0.01 for screening and comparative design, are satisfied after 7 and 13 replications respectively. This has been decided as the best fit according to the most conservative approach. Further plots leading to the same conclusion can be found at [7.9](#) in the appendix.

4.5 Variance reduction techniques

To enhance the accuracy of the estimates, a variance reduction technique was implemented. The principle of antithetic variables, also referred to as complementary random numbers, was chosen here because the principle allows for the exploitation of negative correlation between paired samples. This allows it to “cancel out” some of the random fluctuations inherent to Monte Carlo Simulation.

To implement this method, a couple of adjustments had to be made to the Python code. First of all, the ‘helper.py’ was altered, specifically with the ‘Exponential_distribution()’ function. This distribution

function allows for the arrivals to be generated, using the uniform distribution to generate random input (j_1 , between 0 and 1) for the exponential distribution function output (j_2). This was changed such that the output generated would include both the original output (j_2) and the complementary output (j_{2inv}) generated by the inverse of the generated random variable ($j_{1inv} = 1 - j_1$) in a list [j_2, j_{2inv}].

After this, the 'simulation.py' file was altered to include both 'runOneSimulation()' and 'runOneSimulationAT()' within a single simulation. 'First, 'runOneSimulation()' would be executed first, making use of the 'generatePatients()' that was altered to make use of generated outputs, j_2 , to generate patient arrivals/phone calls. Upon completion of one simulation, as discussed in section 3.4, the variables used are reset, and 'runOneSimulationAT()' is executed using the same seed value as 'runOneSimulation()'. This ensures consistency in the generated random numbers between the two simulations, with the only difference being the utilization of the newly created 'generatePatients2()' function in the second simulation, which utilizes the inverse of ' j_2 ', denoted as ' j_{2inv} '. After completing this simulation, the relevant outputs from 'runOneSimulation()' and 'runOneSimulationAT()' are averaged to yield the new output for one simulation run, marking the successful completion of the antithetic variables variance reduction method.

For further analysis of the complete adjustments in the 'HelperAdjusted.py' and 'simulation2.py' files, the reader can look at the provided Python files.

5 EXPERIMENTAL DESIGN

In this simulation study, there are 132 possible design points. Resulting from the multiplication of all possible design points for each different factor ($132 = \text{factor A} * \text{factor B} * \text{factor C} = 11 * 3 * 4$). With factor A being the amount of urgent slots options, B being the number of strategies and C being the number of different strategies.

The common random numbers technique is employed to ensure that alternative configurations are compared under equivalent experimental conditions. This methodology enhances the reliability of attributing observed performance differences directly to variations in system configuration, rather than to fluctuations in experimental conditions. Critical for the success of common random numbers is that the runs are properly synchronized, meaning that the generated random number should be used for the same purpose in all the different model configurations.

5.1 Screening design

5.1.1 Stage 1: One-factor-at-a-time

A screening design is performed to screen out the important main effects and decrease the number of alternative designs.

A one-factor-at-a-time approach is used with $k = 3$ with k equal to the number of factors. This approach suits the problem because 2 out of 3 factors that influence the performance are qualitative with 3 and 4 levels respectively. If certain levels of these qualitative factors can be screened out, full factorial screening will be used next.

While keeping all other factors fixed, the effect of varying the number of slots was studied first. Configurations with 16 slots and above are removed as this has a significant undesirable effect on the response for all strategies and rules. This can be seen in Figure 7. Furthermore, because there is an upwards leaning trend in the objective value (which is desirable) the choice was made to remove 15

slots as well. Finally, the usage of 11 time slots for urgent patients is also neglected since it has a weaker performance than the usage of 12 time slots, especially for strategy 3. This can be seen in Figure 10 in Appendix 7.10.

The removal of the specific design points has led to a significant reduction in overall design points, already improving the efficiency. Nevertheless, in Figure 7 it is very clear that, for 12 and 14 slots and for every rule, strategy 1 has significantly weaker performance than strategy 2 and 3. Therefore it is appropriate to reject strategy 1 and only take strategy 2 and 3 into further consideration.

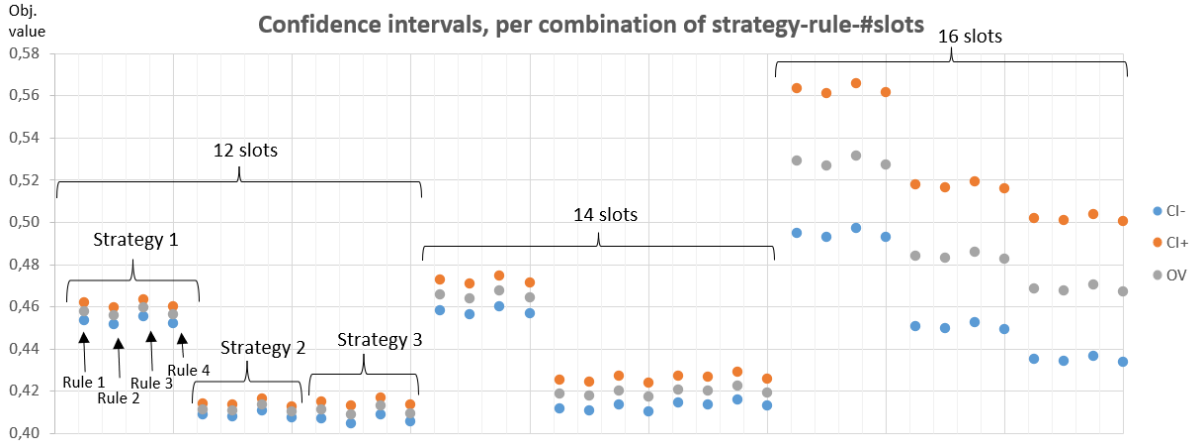


Figure 7: The confidence intervals and objective value per combination of the strategy, the rules and the number of urgent time slots

5.1.2 Stage 2: Full factorial design

For further distinction, the 2^k factorial design with $k = 3$ is used to determine the factor and measure the effects and interactions between the 2 remaining strategies, the 4 rules and the interval [12,14] for slots for urgent patients. There are $2 * 4 * 3 = 24$ design points remaining.

The general model is used:

$$J(\alpha) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_{12} x_1 x_2 + \beta_{13} x_1 x_3 + \beta_{23} x_2 x_3 + \beta_{123} x_1 x_2 x_3$$

The main effects are calculated with the following formula:

$$e_j = \frac{\sum_{l=1}^k a_{jl} R_l}{2^{k-1}} \text{ with } a_{jl} = 1 \text{ ('+' level) and } a_{jl} = -1 \text{ ('-' level)}$$

The interaction effects are calculated with the following formula:

$$e_j = \frac{\sum_{l=1}^k a_{j_1 l} a_{j_2 l} R_l}{2^{k-1}} \text{ with } a_{j_1 l} = 1 \text{ ('+' level) and } a_{j_1 l} = -1 \text{ ('-' level)}$$

The 90% confidence intervals for the statistical significance of effects are calculated as follows:

$$\left[\bar{e}_j(n) - S_j \frac{t_{n-1} \frac{0.1}{2}}{\sqrt{n-2}}, \bar{e}_j(n) + S_j \frac{t_{n-1} \frac{0.1}{2}}{\sqrt{n-2}} \right] \text{ with } n \text{ the number of replications}$$

To reduce the number of design points rule 1 and 2 will be compared, also rule 3 and 4, rather than comparing each of the 4 different rules with each other.

In the following tables, the lower and upper numerical values are shown for the three different factors.

Numerical value	Factor A	Factor B	Factor C
-1	12 slots	Rule 1	Strategy 2
1	14 slots	Rule 2	Strategy 3

Table 1: the numerical values for the three different factors for the comparison between rule 1 and 2

Numerical value	Factor A	Factor B	Factor C
-1	12 slots	Rule 3	Strategy 2
1	14 slots	Rule 4	Strategy 3

Table 2: the numerical values for the three different factors for the comparison between rule 3 and 4

5.1.2.1 Rule 1 and 2

Figure 8, presented below, illustrates the averages derived from the seven replications for both the main and the side effects. The table with values can be found in the appendix at [7.11](#). Upon examination of the graph in Figure 8, it becomes evident that 12 urgent time slots are preferred over the 14 slots. The main effect of factor A, which is positive, refers to the average change in response when moving from 12 to 14 slots. Since the objective is to minimize the objective value, this is suboptimal. However, due to interaction effects of factor A (the number of urgent time slots) and factor C (the strategy), a definitive conclusion cannot be drawn. Nevertheless, for the sake of improving the efficiency, the one-factor-at-a-time approach is considered. The downward trend of the objective value when reducing the number of urgent time slots is seen as more important than the interaction effect. A similar conclusion for the strategy cannot be made since there are no significant effects in the one-factor-at-a-time approach seen in Figure 7 and the appendix at [7.10](#).

For factor B, a negative average effect is shown in Figure 8 which means that it has a negative effect on the objective value on average going from rule 1 to rule 2. Our objective is to minimize the objective value, hence our choice to reject rule 1. The A x B interaction is neglected due to its small effect. The interaction effects from the other factors with factor B are insignificant which makes this a definitive conclusion.

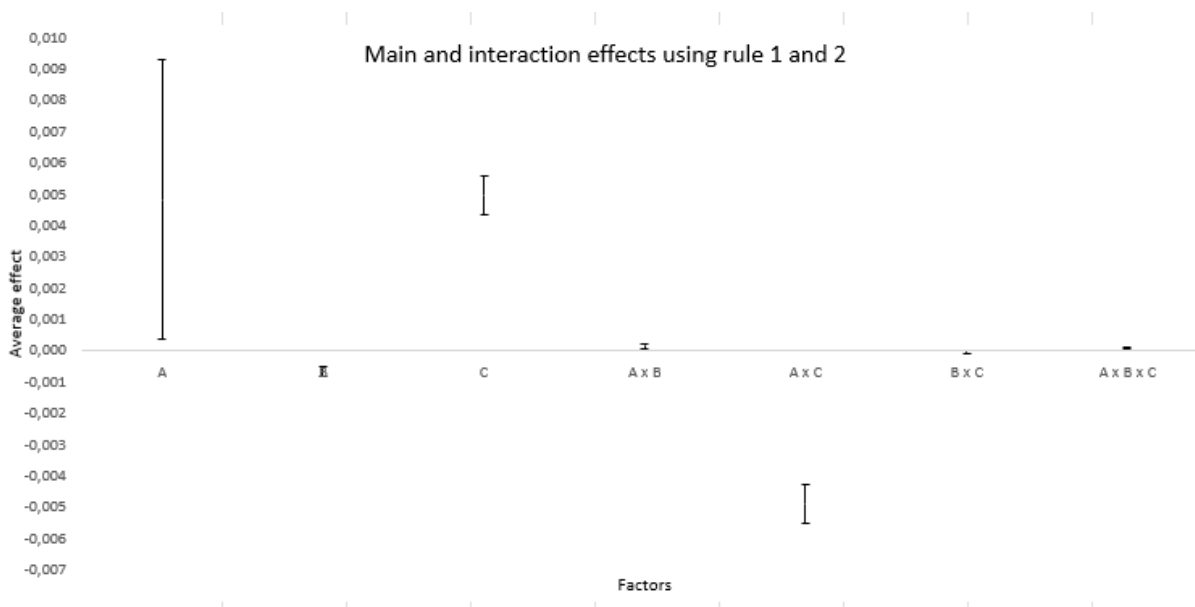


Figure 8: Plot of the averages derived from the 7 replications of the main and interaction effects for rule 1 and 2 (factor B) and the 2 other factors

5.1.2.2 Rule 3 and 4

Now rule 3 and 4 are considered. By looking at factor A in Figure 9, the usage of 12 urgent time slots has a higher preference than the usage of 14 slots. Moving from slots to 14 slots increases on average the objective value, which is undesirable. Again, due to interaction effects of factor A (the number of urgent time slots) and factor C (the strategy), a definitive conclusion cannot be drawn. By looking at the one-factor-at-a-time approach, the reduction in the number of urgent time slots consistently leads to a decrease in the objective value leading to the removal of 14 slots. As mentioned before, for the strategy, a similar conclusion cannot be made since there is no visible difference in the one-factor-at-a-time approach.

For factor B, a negative on average effect is shown which means that moving from rule 3 to rule 4 has on average a negative effect on the objective value. This is desirable, hence the rejection of rule 3. The absence of interaction effects with factor B makes this a definitive conclusion.

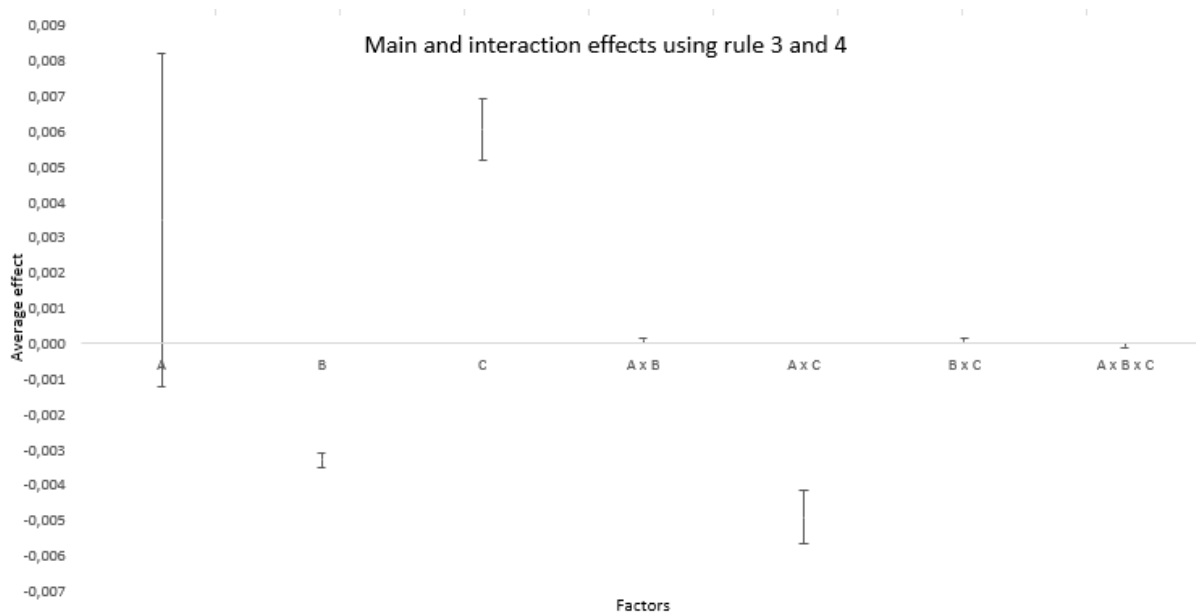


Figure 9: Plot of the averages derived from the 7 replications of the main and interaction effects for rule 3 and 4 (factor B) and the 2 other factors

5.2 Comparison design

The screening design showed that all three factors have a significant influence, meaning that all three should still be considered. Using a sequence of comparative designs, the aim is to provide the hospital with good recommendations on which combination of values to use for the three factors. The screening design showed that rule 2 and 4 and strategy 2 and 3 are worth further investigation, for both 12 and 13 slots. This leads to 8 design points, which can be found in the table below.

Design point	#Slots	Rule	Strategy
X1j	12	2	2
X2j	12	2	3
X3j	12	4	2
X4j	12	4	3
X5j	13	2	2
X6j	13	2	3
X7j	13	4	2
X8j	13	4	3

Table 3: Design points

5.2.1 All pairwise comparison design

The decision was made to utilize an all pairwise comparison model to assess and contrast all remaining design points, as opposed to solely comparing them with a single standard design (for example comparing with the current design). This approach offers distinct benefits, especially in situations demanding a thorough comprehension of the relative performance of each option, rather than merely grasping the overall response landscape depicted by response surfaces (Montgomery, 2012). In the previous paragraph, it was mentioned that there are eight design points. If an all-pairwise comparison design is adopted, every possible pair of design points is compared, which will lead to $k(k-1)/2$ confidence intervals, in this case $8(8-1)/2 = 28$. Thus, 28 different combinations are possible, and this seems an acceptable number to go for the all-pairwise comparison design. Each pair is compared by using a paired t-test according to the following formula: $\bar{d} \pm t^* * \frac{s_d}{\sqrt{n}}$. The great advantage of such an all-pairwise comparison design is that the interpretation is very easy, and thus it is clear to see when a certain design point is significantly better than another design point. For the calculations the number of runs is set equal to 13, as this was decided in part 4.4. In order to establish a confidence level of at least $1-0.1 (=0.9)$, each individual interval should be constructed at the level $1 - 0.1/\frac{8*(8-1)}{2}$ (approximately = 0.99643). With the degrees of freedom equal to 12, the t-value equals 3.611. The confidence intervals can be found in Appendix [7.12](#).

[Table F](#) indicates that design point 7 exhibits significant superiority compared to all other design points, with the exceptions of design points 1 and 3. However, for pairs 1 vs 7 and 3 vs 7, the confidence intervals encompass zero, thereby preventing rejection of the null hypothesis, which suggests equal performance between these design points. Upon comparison between design points 1 and 3, the latter demonstrates superior performance. Consequently, design point 1 is eliminated from consideration, and the analysis proceeds with design points 3 and 7, employing Dudewicz and Dalal's two-stage sampling procedure. Through this ranking and selection procedure, the goal is to reach a definitive conclusion.

5.2.2 Two-Stage sampling procedure of Dudewicz and Dalal

5.2.2.1 First stage

For this method, the same 13 replications from the all pairwise comparison design are utilized. The total number of runs necessary for both systems to reach a decision needs to be computed using the following formula:

$$N_i = \max \{n_0 + 1, \left\lceil \frac{h_1^2 S_i(n_0)}{(d^*)^2} \right\rceil\}$$

The value for h_1 is a constant that depends on k , n_0 and P^* and can normally be obtained from the Dudewicz and Dalal's table:

P^*	n_0	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$	$k=10$
0.90	20	1.896	2.342	2.583	2.747	2.870	2.969	3.051	3.121	3.182
0.90	40	1.852	2.283	2.514	2.669	2.785	2.878	2.954	3.019	3.076
0.95	20	2.453	2.872	3.101	3.258	3.377	3.472	3.551	3.619	3.679
0.95	40	2.386	2.786	3.003	3.150	3.260	3.349	3.422	3.484	3.539

Table 4: Segment of Dudewicz and Dalal's table

P^* remains at 0.9, where both n_0 and k are respectively equal to 13 and 2 (number of design points). d^* is set at 0.000633196, a value 0.00003 less than the difference between X_3 and X_7 . In this table, the scenario where n_0 is 0 is not included. If h_1 were estimated to be approximately 1.896 (since 13 is

the closest value to 20), this estimation would result in high values for N_i , as illustrated in the table below.

First stage			
i	$X_i(13)$	$S_i^2(13)$	N_i
3	0.40935079	1.6429E-05	148
7	0.4086876	2.7372E-05	246

Table 5: Results of first stage sampling procedure

According to Dudewicz and Dalal's table, if the assumed n_0 would be adjusted from 20 to 13 (moving closer to the true value), h_1 would consequently increase. This would lead to even higher values for N_i .

5.2.2.2 Second stage

To initiate the second stage, additional $N_i - 13$ runs of system i (where $i = 1$ and 2) are required. However, conducting such a large number of runs could be time-consuming and inefficient. In combination with the uncertainty surrounding the true value of h_1 , it is advisable not to proceed with this method. Instead, to still arrive at a final recommendation, the secondary objectives can be utilized to decide between design points 3 and 7.

5.2.3 Comparing by secondary objectives

Design points 3 and 7 both consist of rule 4 and strategy 2. Therefore, the secondary objectives will exclusively determine the choice between implementing 12 or 13 slots. Paired t-tests have been performed separately for each secondary objective: once for elective scan waiting times and once for comparing the overtime. Once again, 13 runs are conducted, with degrees of freedom equal to 12, and confidence levels chosen at 0.90. For both secondary objectives, design point 7 outperforms design point 3. This result indicates that design point 7 is the overall best in terms of both the objective function and elective scan waiting times, as well as overtime, since the confidence intervals are positioned to the left of zero.

t-value	Mean ESWT X7-X3	Half length	CL-	CL+
1.782288	-0.000527385	9.64797E-05	-0.00062	-0.00043

Table 6: Results confidence intervals for difference in elective scan waiting times between design 3 and 7

t-value	Mean OT X7-X3	Half length	CL-	CL+
1.782288	-0.013612846	0.000551769	-0.01416	-0.01306

Table 7: Results confidence intervals for difference in overtime between design 3 and 7

6 RESULTS

6.1 Final result

The simulation study ends with the selection of 1 design out of 132 total possible design combinations. A design with 13 urgent slots, using the urgent slots timing strategy (strategy 2), complemented with using scheduling rule 4 performs best.

6.2 Limitations

One factor at a time approach is useful for the initial screening but has its limitations. By only varying one factor at a time, the interaction between distinct factors might not be observed. This may lead to a lower performance of the design. Furthermore, the approach relies on the assumption that the factors are varying independently and influencing the response variables.

Another limiting factor is the uncertainty surrounding the assumptions that have been made, here below these are listed:

- Input data assumptions
- Simulation model constraints
- Number of simulation runs and variance reduction techniques
- Experimental design
- Statistical methods and interpretation

These limitations suggest that while the simulation provides valuable insights into the scheduling system's performance under specified conditions, the results should be interpreted with caution. The assumptions and simplifications necessary for computational feasibility might limit the applicability of the findings to real-world scenarios where system behavior could be more complex and less predictable.

At last, there are many more time slots, scheduling rules and distribution strategies but in this simulation study the department has limited it to the number of urgent slots, rules and strategies mentioned above. This means that there could exist a better solution out of the scope of this analysis.

6.3 Conclusions

This report forms the analysis on the construction of a cyclic appointment schedule for an outpatient radiology department with a single server. The goal of this study is to minimize the waiting times for urgent and elective patients while considering the impact on the daily overtime while considering 3 different factors: number of slots, rule, and the chosen strategy. The problem underwent appropriate simulation and optimization techniques to identify the best alternative. The entire evaluation resulted in the rejection of all the null hypotheses, as stated in section 1.3.2.

Following the calculations from the screening design, it resulted in 8 relevant design points. After performing comparison design, it was determined that design point 7 emerges as the optimal choice. This configuration features 13 slots, rule 4 and strategy 2. It provides the most efficient solution for minimizing waiting times for both urgent and elective patients.

7 APPENDIX

7.1 Alternative rules explained

7.1.1 Bailey-Welch rule

The Bailey-Welch rule involves assigning the first K patient to the same time slot, in this project the K is set equal to 2. For example, the first patient is assigned to slot 10 that starts at 10:15. The next patient is assigned slot 11 that starts at 10:30. Both patients 1 and 2 will then have an appointment time at 10:15. Patient 3 assigned slot 12 (starts at 10:45) will have an appointment time of 10:30, just like patient 4 who is assigned to slot 13.

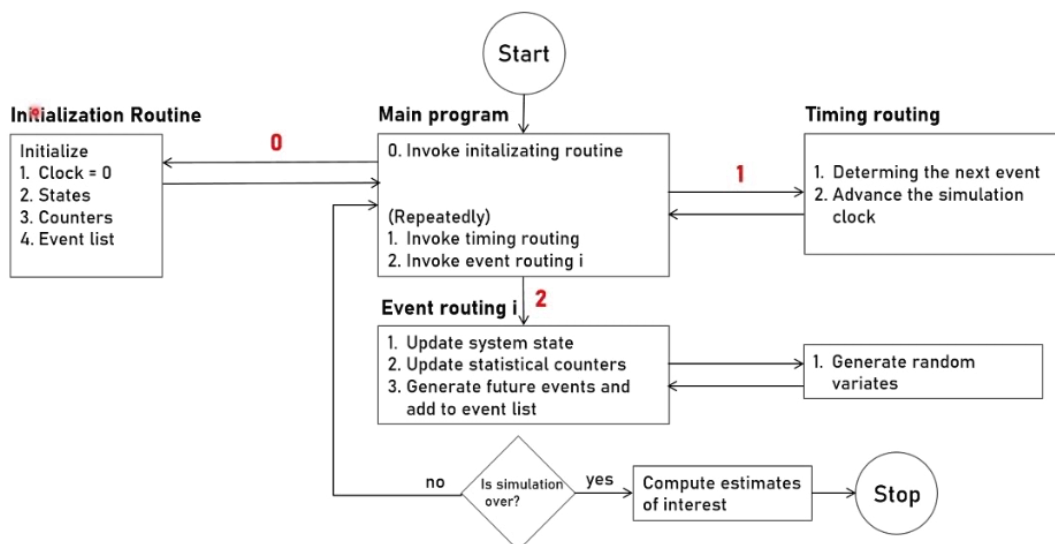
7.1.2 Blocking rule

The blocking rule demands selecting a value for B , with B equal to the number of slots per block. In this case with $B=2$, two time slots are combined and the patients in both slots are also assigned the same appointment time. So, the patients of slot 1 and 2, which start at 08:00 and 08:15 respectively, will have an appointment time of 08:00.

7.1.3 Benchmarking rule

The benchmarking rule dictates setting the appointment time equal to the start time of the slot that the patient is assigned to, minus $k_\sigma * \sigma_e$, with σ_e representing the standard deviation of the scan duration of elective patients and k_σ , here 0.5, showing how much the possible deviations in the scan duration are taken into account.

7.2 Discrete Event Simulation diagram flow

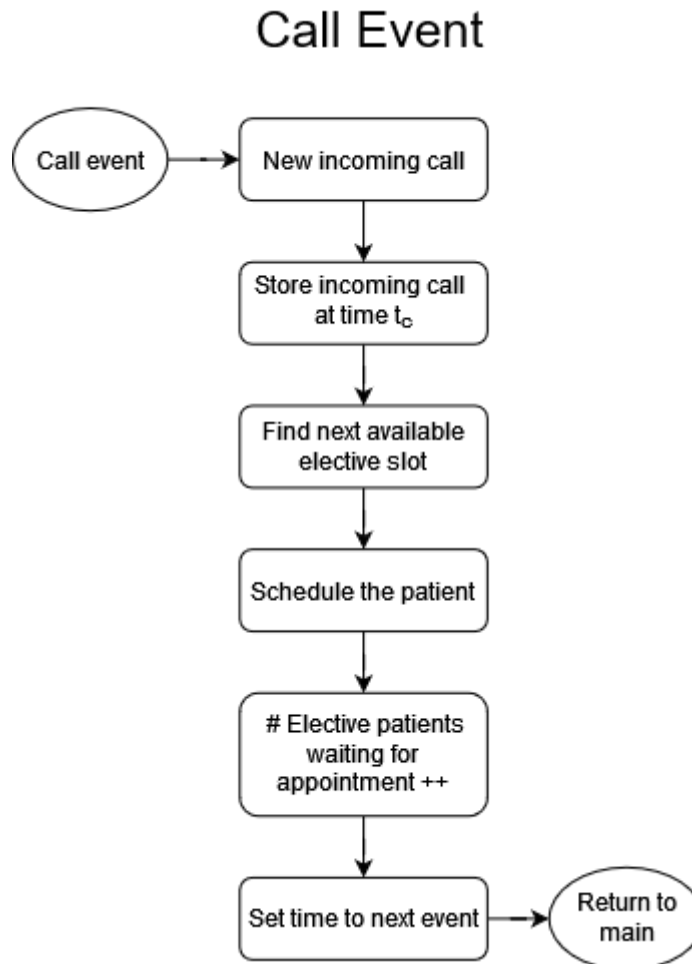


Source: Law

(A Gentle Introduction to Discrete-event Simulation, 2024)

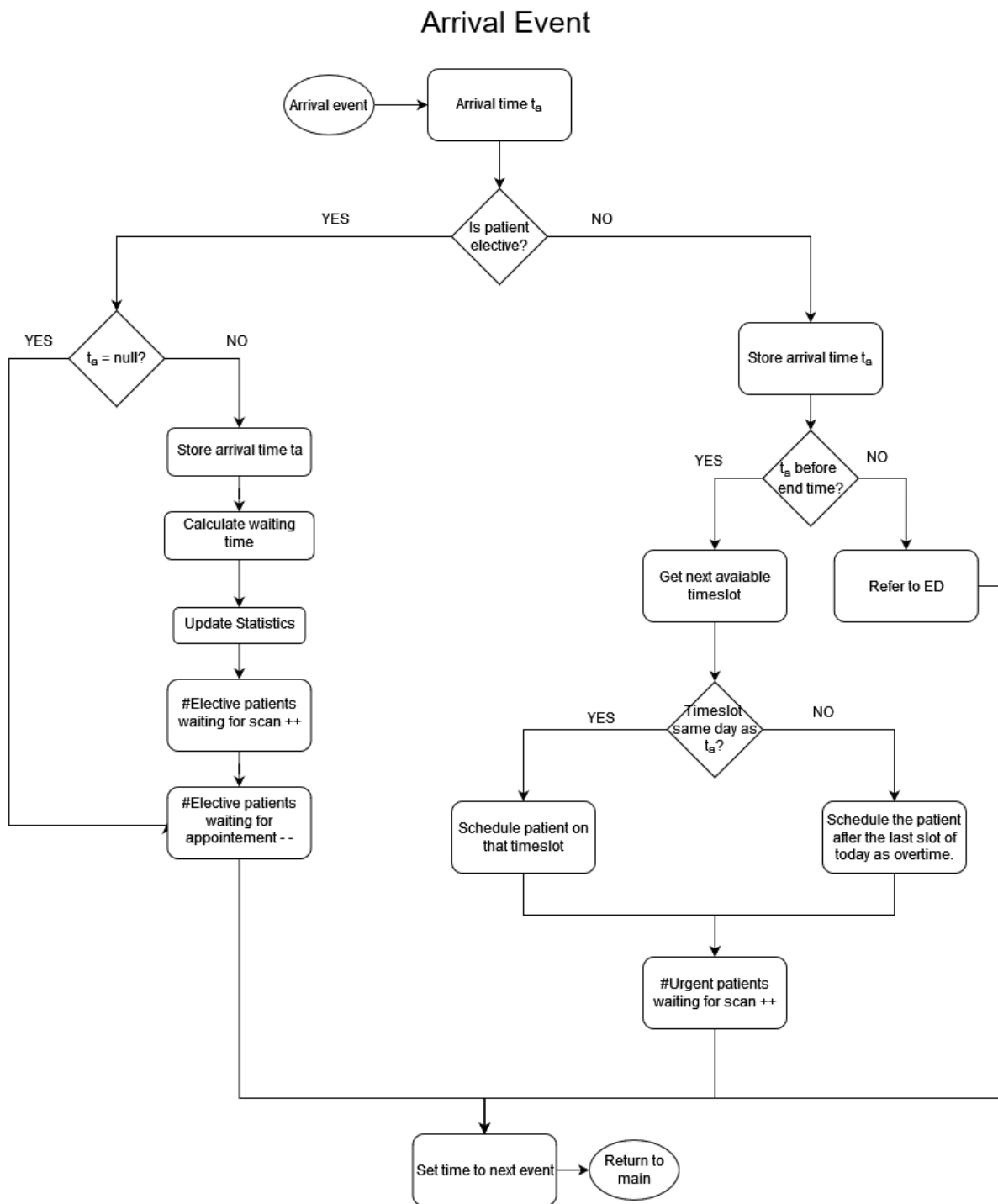
7.3 Flowcharts

7.3.1 Call event



Flowchart 1: call event

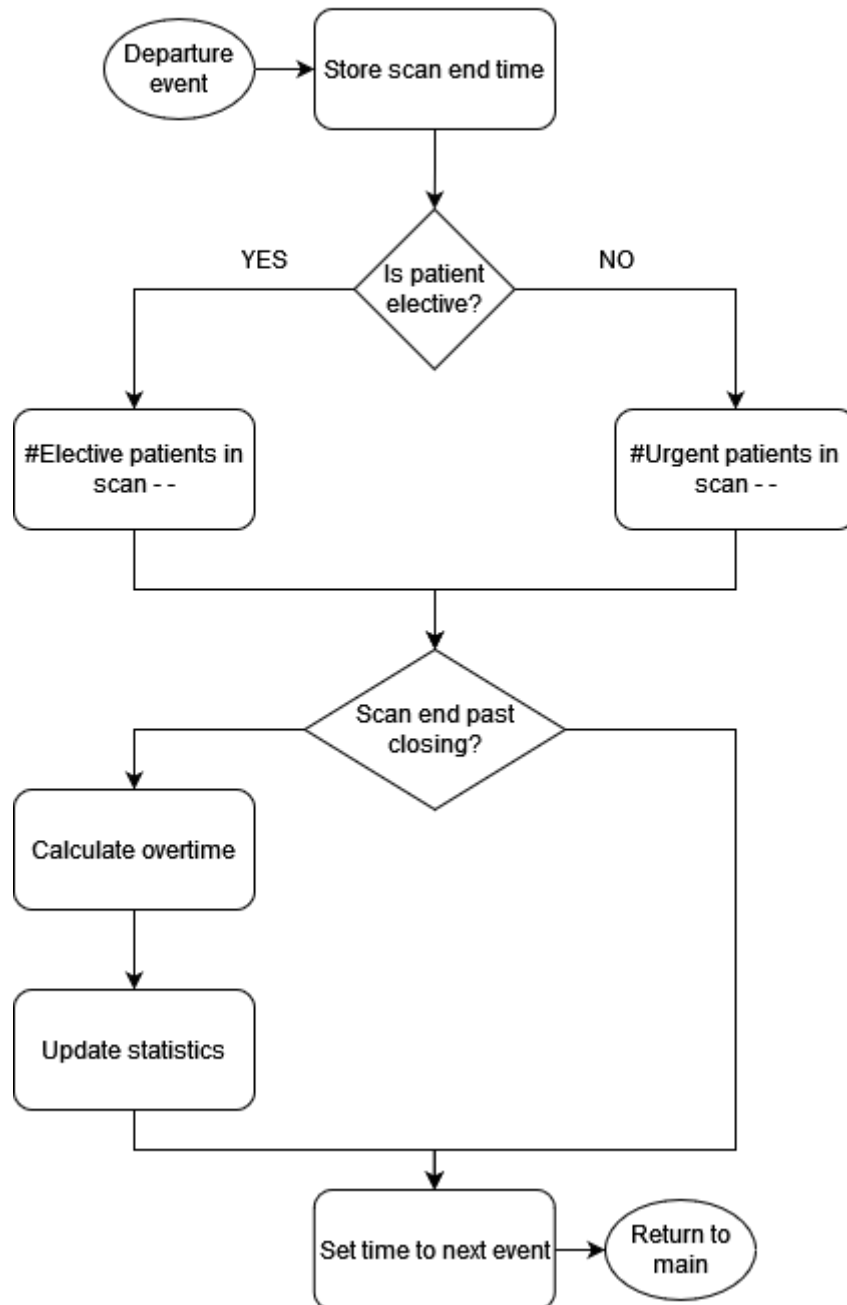
7.3.2 Arrival event



Flowchart 2: arrival event

7.3.3 Departure event

Departure Event



Flowchart 3: departure event

7.4 Table A: Input Data

Input Data	Further Explanation
Total amount of timeslots	160 timeslots are available in total
Maximum waiting time	For $P_u = 9$ hours, for $P_e = 168$ hours
Operational variability	Variability related to waiting time and service time
Arrival rate / variability	$P_e \sim \text{Poisson}(\lambda = 28.345)$ $P_u \sim \text{neg exp}(\lambda = 2.5)$ full day open $P_u \sim \text{neg exp}(\lambda = 1.25)$ half day open
Tardiness	$P_e \sim N(0, 2.5^2)$, no show = 2%
Appointment waiting time	Elective patients must wait for an appointment
Scan duration	$P_e \sim N(15, 3^2)$ $P_u \sim$ see table B below $(\mu = 17.5; \sigma = 2.375)$
Unpunctuality	Difference arrival time and appointment time
Overtime	Positive difference in time between the scan end time of the last patient on a particular day and the session end time of the day

Decision Variables	Further Explanation
Capacity distribution between P_u and P_e	Number of timeslots dedicated to the urgent patients $\in [10;20]$
Strategy for planning horizon	3 different strategies to choose from
Appointment scheduling rules	4 different rules to choose from

Table A: Input data & decision variables

7.5 Table B: Frequency & scan duration distribution for urgent patients

Scan Type	Freq. (%)	Scan duration	
		Mean (min)	St dev (min)
Brain	70%	15	2.5
Spine			
Lumbar vert	10%	17.5	1
Cervical ver	10%	22.5	2.5
Abdomen MI	5%	30	1
Others	5%	30	4.5

Table B: Frequency & scan duration distribution for urgent patients

7.6 Table C: Combined plots running averages strategy 1

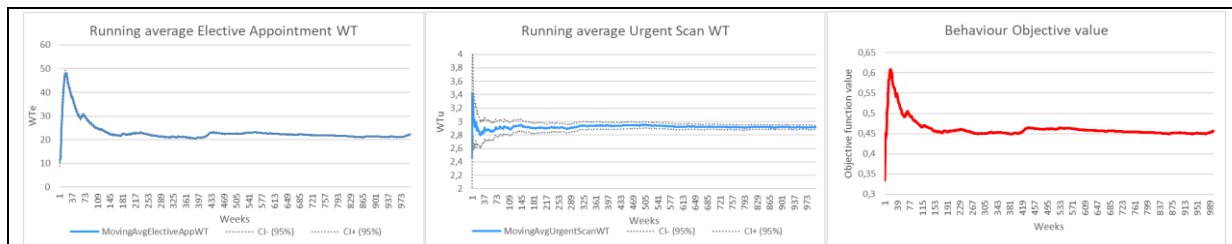


Figure C1: combined plot for the running averages (strategy 1, **rule 1**, 14 slots) of the elective appointment waiting time (left), the urgent scan waiting time (center) and the objective function value (right)

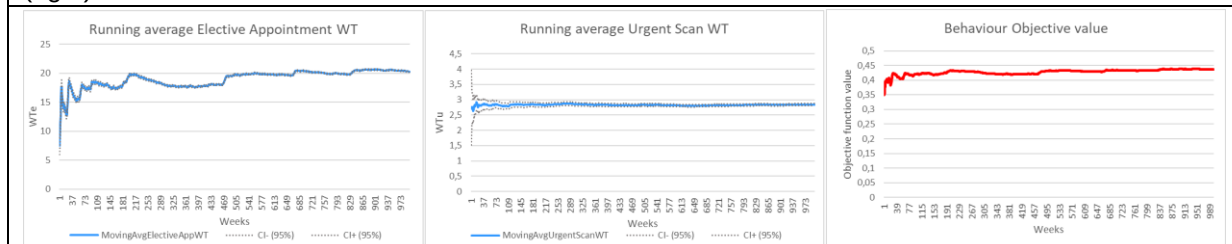


Figure C2: combined plot for the running averages (strategy 1, **rule 2**, 14 slots) of the elective appointment waiting time (left), the urgent scan waiting time (center) and the objective function value (right)

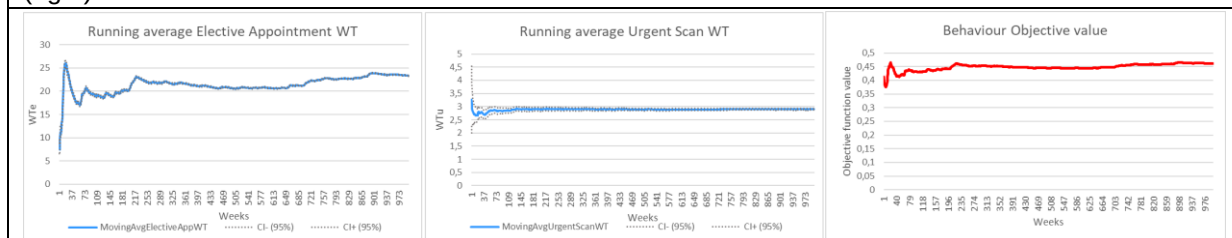


Figure C3: combined plot for the running averages (strategy 1, **rule 3**, 14 slots) of the elective appointment waiting time (left), the urgent scan waiting time (center) and the objective function value (right)

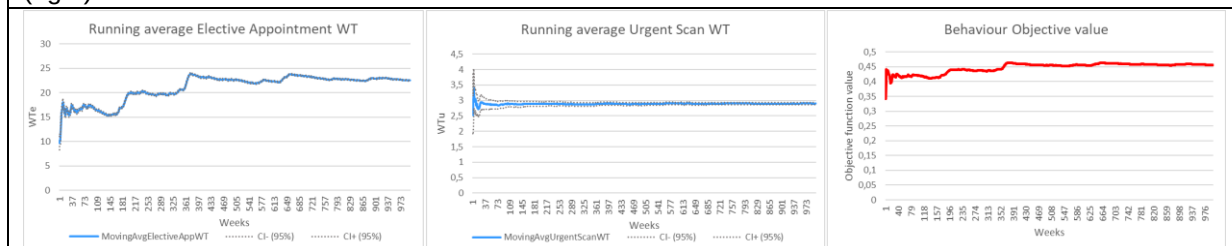


Figure C4: combined plot for the running averages (strategy 1, **rule 4**, 14 slots) of the elective appointment waiting time (left), the urgent scan waiting time (center) and the objective function value (right)

7.7 Table D: Combined plots running average for strategy 2 & 3 – rule1 - 14 slots

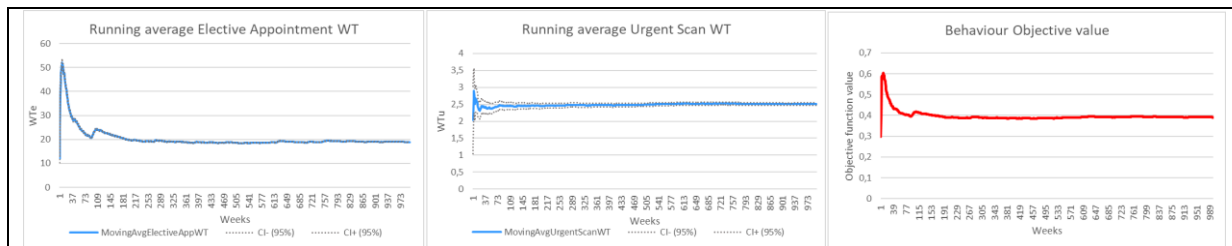


Figure D1: combined plot for the running averages (**strategy 2**, rule 1, 14 slots) of the elective appointment waiting time (left), the urgent scan waiting time (center) and the objective function value (right)

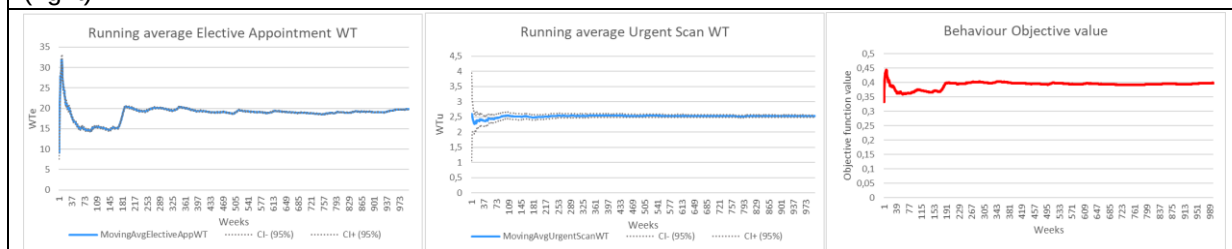


Figure D2: combined plot for the running averages (**strategy 3**, rule 1, 14 slots) of the elective appointment waiting time (left), the urgent scan waiting time (center) and the objective function value (right)

7.8 Table E: Combined plots running average for 10, 14 & 16 slots – strategy 1 – rule 2

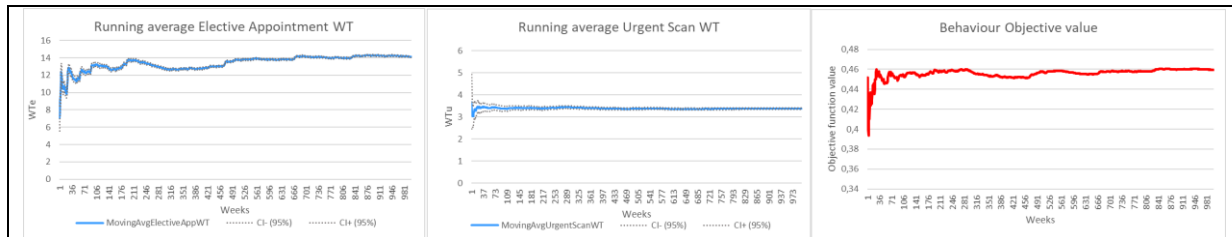


Figure E1: combined plot for the running averages (strategy 1, rule 2, **10 slots**) of the elective appointment waiting time (left), the urgent scan waiting time (center) and the objective function value (right)

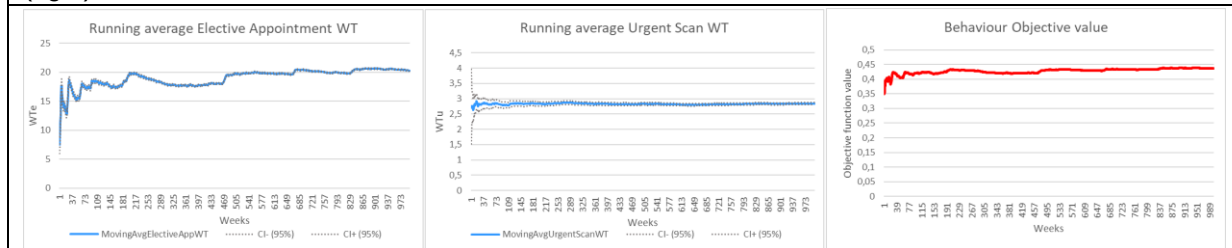


Figure E2: combined plot for the running averages (strategy 1, rule 2, **14 slots**) of the elective appointment waiting time (left), the urgent scan waiting time (center) and the objective function value (right)

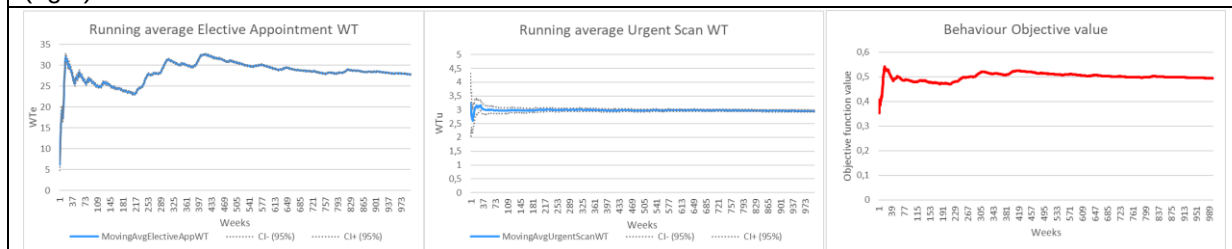
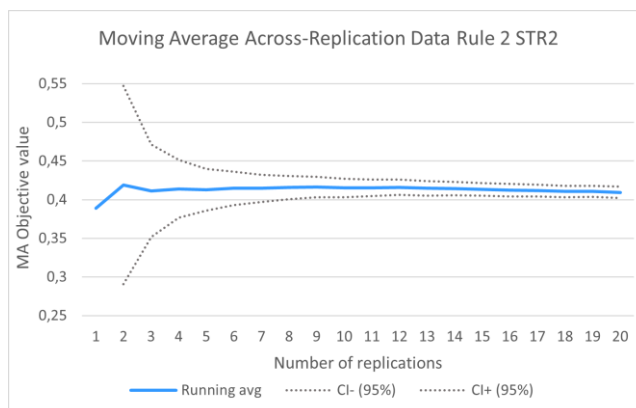
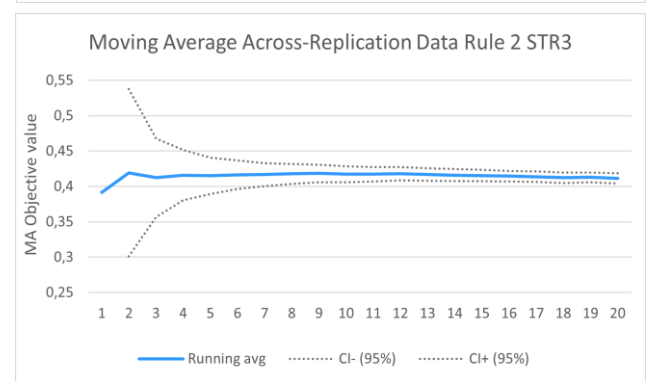
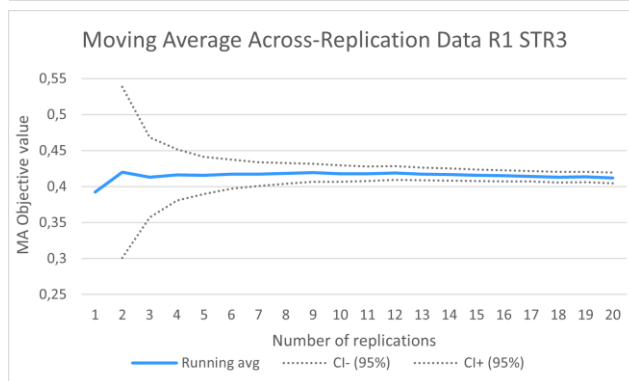
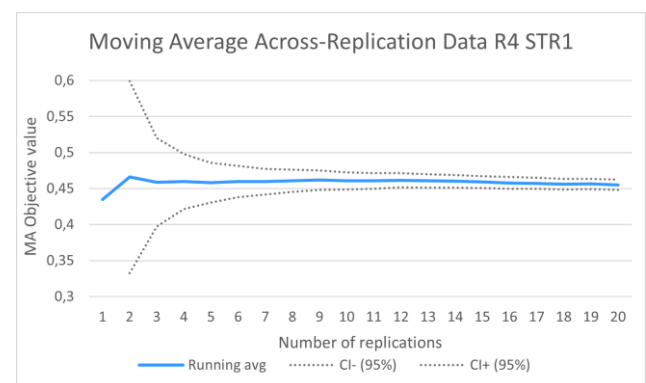
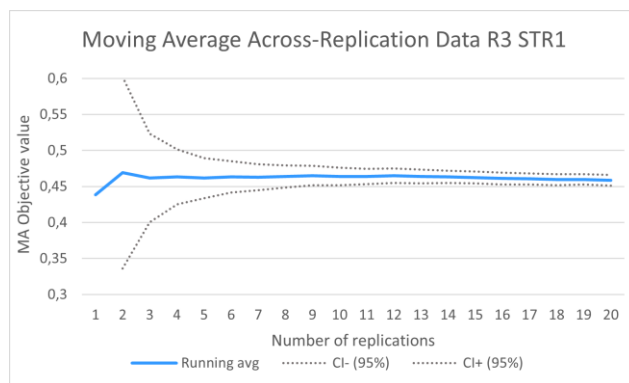
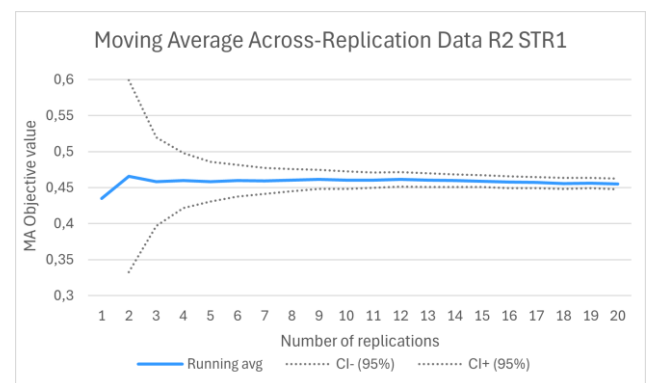
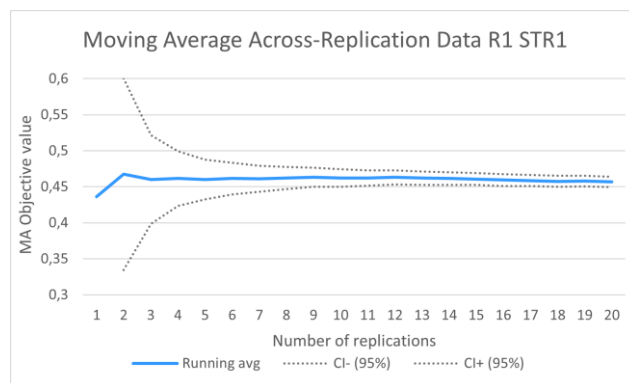


Figure E3: combined plot for the running averages (strategy 1, rule 2, **16 slots**) of the elective appointment waiting time (left), the urgent scan waiting time (center) and the objective function value (right)

7.9 Moving averages across-replication data 14 slots



Some examples are shown here of the across-replication data. On the vertical axis the moving average of the objective value can be found. The horizontal axis shows the number of replications. This visualizes that generally the half width of 0.02 and 0.01 for screening and comparative design, are satisfied after 7 and 13 replications respectively.

7.10 The confidence intervals and objective value per combination of the strategy, the rules and the number of urgent time slots

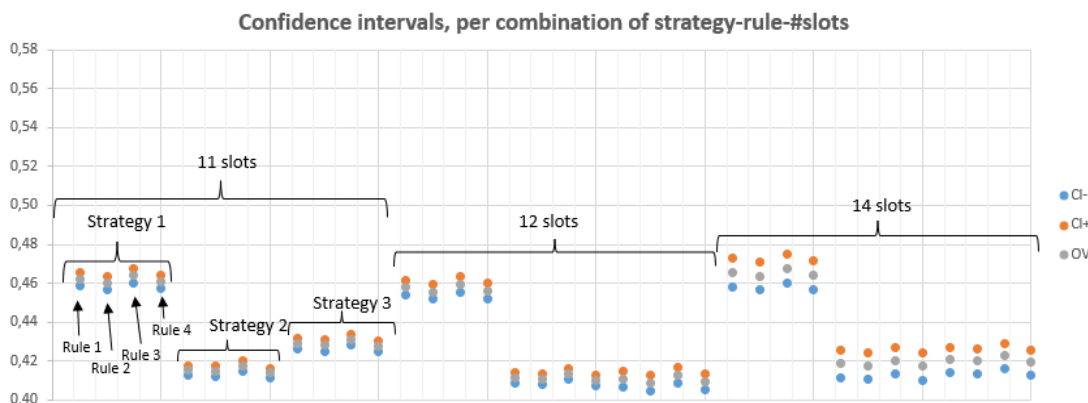


Figure 10: The confidence intervals and objective value per combination of the strategy, the rules and the number of urgent time slots given in the figure.

7.11 Values for the main and interaction effects in the screening design

Rule 1 and 2

	Avg	St Dev	CI-	CI+	X-axis label	Half-width
Main effect A	0,0048	0,0060	-0,0004	0,0101	A	0,004439217
Main effect B	-0,0007	0,0002	-0,0009	-0,0005	B	0,000150409
Main effect C	0,0049	0,0008	0,0042	0,0057	C	0,000616327
Interaction A x B	0,0001	0,0001	0,0000	0,0002	A x B	8,47969E-05
Interaction A x C	-0,0049	0,0008	-0,0057	-0,0042	A x C	0,000616327
Interaction B x C	-0,0001	0,0001	-0,0001	0,0000	B x C	4,70218E-05
Interaction A x B x C	0,0001	0,0001	0,0000	0,0001	A x B x C	4,70218E-05

Rule 3 and 4

	Avg	St Dev	CI-	CI+	X-axis label	Half-width
Main effect A	0,0035	0,0064	-0,0012	0,0091	A	0,004713534
Main effect B	-0,0033	0,0003	-0,0035	-0,0031	B	0,000194731
Main effect C	0,0061	0,0012	0,0052	0,0071	C	0,000857461
Interaction A x B	0,0001	0,0001	0,0000	0,0002	A x B	6,57738E-05
Interaction A x C	-0,0049	0,0010	-0,0057	-0,0040	A x C	0,000746537
Interaction B x C	0,0001	0,0001	0,0000	0,0002	B x C	6,6543E-05
Interaction A x B x C	-0,0001	0,0001	-0,0001	0,0000	A x B x C	7,04748E-05

7.12 Table F: All-pairwise comparison: confidence intervals

$i_1 \backslash i_2$	2	3	4	5	6	7	8
1	0.00933±0.00168*	-0.00072±0.00022*	0.00896±0.00164*	-0.00076±0.00179	0.00583±0.00205*	-0.00138±0.00185	0.00531±0.00197*
2		-0.01005±0.00175*	-0.00037±0.00022*	-0.01009±0.00231*	-0.0035±0.0017*	-0.01071±0.00243*	-0.00402±0.00168*
3			0.00968±0.00171*	-0.00003±0.00174	0.00655±0.00208*	-0.00066±0.00178	0.00603±0.00199*
4				-0.00971±0.00234*	-0.00313±0.00181*	-0.01034±0.00245*	-0.00365±0.00177*
5					0.00658±0.00137*	-0.00063±0.00022*	0.00606±0.00131*
6						-0.00721±0.00148*	-0.00052±0.0002*
7							0.00669±0.0014*

Table F: Confidence intervals of the all-pairwise comparison

7.13 Excel files

7.13.1 Length of Simulation

7.13.2 Method of Welch

7.13.3 outputAcross

7.13.4 Screening Design

7.13.5 Full factorial screening

7.13.6 Comparison Design

7.14 Code files

7.14.1 Simulation 1

7.14.2 Simulation 2

7.14.3 Simulation 3

7.14.4 HelperAdjusted

8 REFERENCES

1. Robinson, S. (2004). *Simulation: The Practice of Model Development and Use*. John Wiley & Sons.
2. *A gentle introduction to discrete-event simulation*. (2024, January 6). A Gentle Introduction to Discrete-event Simulation. <https://softwaresim.com/blog/a-gentle-introduction-to-discrete-event-simulation/>
3. Montgomery, D. C. (2012). *Design and Analysis of Experiments*. Wiley.